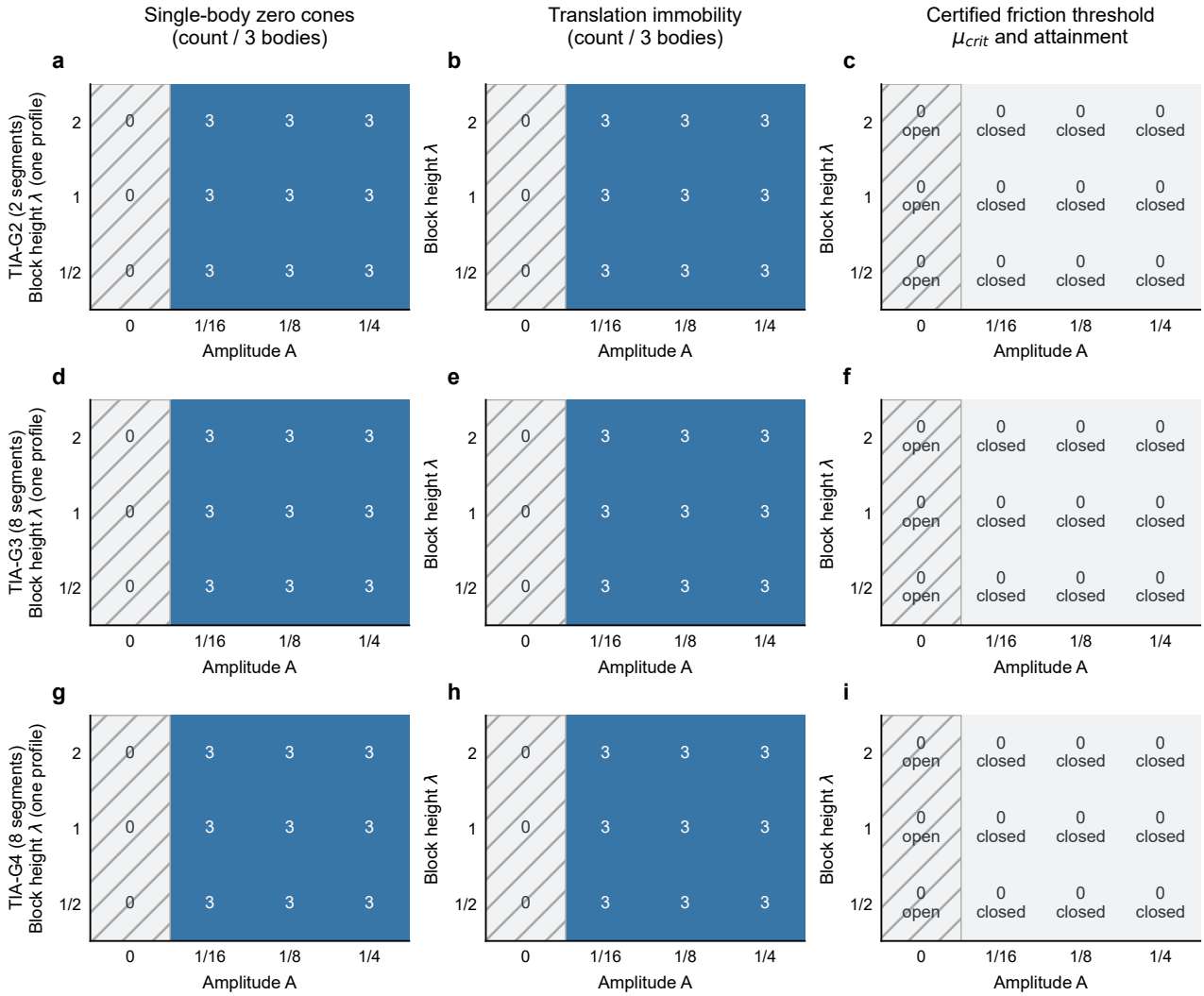

Hatched A = 0: skeleton change. All thresholds are zero; no shape-design trend is established.

2 / 8 / 8-segment polygons only: circular-grid refinement can retain false smooth-interface locking.

Associated-flow positive thresholds are unsafe-side lower bounds; no smooth-curve convergence is certified.